Lecture 9

Adjoints; Self-Adjoint, Unitary, and Normal Operators

The Riesz theorem of Lecture 8 turned vectors into functionals. Applied to operators it produces the *adjoint* $T^\dagger$, the abstract version of the conjugate transpose, defined by $\langle T^\dagger u, v\rangle = \langle u, Tv\rangle$. With the adjoint in hand we can single out the three families of operators that organize all of quantum mechanics: the *self-adjoint* operators $T^\dagger = T$, whose real eigenvalues are the possible outcomes of a measurement; the *unitary* operators $U^\dagger = U^{-1}$, which preserve inner products and so implement symmetries and time evolution; and the *normal* operators $T^\dagger T = TT^\dagger$, the widest class that admits an orthonormal eigenbasis. This lecture builds the adjoint and proves the spectral facts—real eigenvalues, orthogonal eigenvectors, unit-circle spectra—that the spectral theorem of Lecture 10 will complete.

Learning objectives

By the end of this lecture you will be able to:

- Compute the adjoint $T^\dagger$ and use $\langle T^\dagger u, v\rangle = \langle u, Tv\rangle$ and its algebra.
- Classify operators as self-adjoint, unitary, or normal.
- Prove the core facts: real eigenvalues, unit-circle spectrum, orthogonal eigenvectors.
- Identify observables, symmetries, and the class the spectral theorem diagonalizes.

1 The adjoint

Definition 9.1 (Adjoint). Let $T \in \mathcal{L}(V)$ on a finite-dimensional inner product space. The *adjoint* $T^\dagger$ is the unique operator satisfying

$$\langle T^\dagger u, v\rangle = \langle u, Tv\rangle \qquad \text{for all } u, v \in V.$$

Existence and uniqueness are Riesz in disguise: for fixed u, the map $v \mapsto \langle u, Tv\rangle$ is a linear functional, so equals $\langle w, v\rangle$ for a unique w; setting $T^\dagger u = w$ defines a linear operator. Equivalently, taking conjugates, $\langle Tu, v\rangle = \langle u, T^\dagger v\rangle$: the adjoint moves an operator across the inner product from one slot to the other.

Proposition 9.2 (The adjoint is the conjugate transpose). In an orthonormal basis, $\mathcal{M}(T^\dagger) = \mathcal{M}(T)^\dagger$: the matrix of the adjoint is the conjugate transpose of the matrix of T.

Proof. With orthonormal $e_1, \dots, e_n$, the (j,k) entry of T is $T_{jk} = \langle e_j, Te_k\rangle$. Hence

$$(T^\dagger)_{jk} = \langle e_j, T^\dagger e_k\rangle = \langle Te_j, e_k\rangle = \overline{\langle e_k, Te_j\rangle} = \overline{T_{kj}},$$

which is exactly the conjugate transpose. ■

Proposition 9.3 (Algebra of adjoints). For $S, T \in \mathcal{L}(V)$ and $a \in \mathbb{C}$: $(S+T)^\dagger = S^\dagger + T^\dagger$, $(aT)^\dagger = \overline{a}\,T^\dagger$, $(ST)^\dagger = T^\dagger S^\dagger$, $(T^\dagger)^\dagger = T$, and $I^\dagger = I$. If T is invertible then $(T^{-1})^\dagger = (T^\dagger)^{-1}$.

Proof. Each follows by checking the defining relation. For the product, $\langle (ST)^\dagger u, v\rangle = \langle u, STv\rangle = \langle S^\dagger u, Tv\rangle = \langle T^\dagger S^\dagger u, v\rangle$ for all v, so $(ST)^\dagger = T^\dagger S^\dagger$. The conjugate on a comes from conjugate-linearity in the first slot. ■

Theorem 9.4 (Null space and range). For $T \in \mathcal{L}(V)$, $\operatorname{null} T^\dagger = (\operatorname{range} T)^\perp$ and $\operatorname{range} T^\dagger = (\operatorname{null} T)^\perp$.

Proof. $u \in \operatorname{null} T^\dagger$ iff $T^\dagger u = 0$ iff $\langle T^\dagger u, v\rangle = 0$ for all v iff $\langle u, Tv\rangle = 0$ for all v iff $u \perp \operatorname{range} T$. That is the first identity; taking $\perp$ and using $(T^\dagger)^\dagger = T$ gives the second. ■

Example 9.5 (An adjoint by conjugate transpose). $A = \left(\begin{smallmatrix} 1 & 2-i \\ 3i & 4 \end{smallmatrix}\right)$ has adjoint $A^\dagger = \left(\begin{smallmatrix} 1 & -3i \\ 2+i & 4 \end{smallmatrix}\right)$: transpose, then conjugate. Since $A^\dagger \neq A$ and $A^\dagger \neq A^{-1}$, this operator is neither self-adjoint nor unitary. ◁

Example 9.6 (Adjoint of the shift). The right shift $T(z_1, \ldots, z_n) = (0, z_1, \ldots, z_{n-1})$ on $\mathbb{C}^n$ has adjoint the left shift $T^\dagger(z_1, \ldots, z_n) = (z_2, \ldots, z_n, 0)$: from $\langle Tu, v\rangle = \sum_k \overline{u_{k-1}} v_k = \sum_j \overline{u_j} v_{j+1} = \langle u, T^\dagger v\rangle$ the adjoint reverses the shift direction. ◁

Example 9.7 (Adjoint of a product reverses order). For $S = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ and $T = \left(\begin{smallmatrix} 1 & 0 \\ i & 0 \end{smallmatrix}\right)$, $ST = \left(\begin{smallmatrix} i & 0 \\ 0 & 0 \end{smallmatrix}\right)$, so $(ST)^\dagger = \left(\begin{smallmatrix} -i & 0 \\ 0 & 0 \end{smallmatrix}\right) = T^\dagger S^\dagger$. Adjunction reverses the order of a product, just as inversion does. ◁

Example 9.8 (The defining relation, checked). With $T = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, $u = (1, 0)$, $v = (0, 1)$: $\langle u, Tv\rangle = \langle (1, 0), (1, 0)\rangle = 1$, while $T^\dagger u = (0, 1)$ gives $\langle T^\dagger u, v\rangle = \langle (0, 1), (0, 1)\rangle = 1$. The two sides of $\langle T^\dagger u, v\rangle = \langle u, Tv\rangle$ agree, as the definition requires. ◁

Remark 9.9 (Detecting the zero operator). $T = 0 \iff T^\dagger = 0 \iff T^\dagger T = 0$. The last equivalence is handy because $T^\dagger T$ is self-adjoint with $\langle v, T^\dagger T v\rangle = \|Tv\|^2$, so $T^\dagger T = 0$ forces $Tv = 0$ for every v.

Corollary 9.10 (Injectivity and surjectivity). T is injective iff $T^\dagger$ is surjective, and T is surjective iff $T^\dagger$ is injective; hence T is invertible iff $T^\dagger$ is.

Proof. T injective $\iff$ $\operatorname{null} T = \{0\}$ $\iff$ $(\operatorname{range} T^\dagger)^\perp = \{0\}$ $\iff$ $\operatorname{range} T^\dagger = V$ by Theorem 9.4, i.e. $T^\dagger$ surjective. Exchanging T and $T^\dagger$ gives the other statement. ■

2 Self-adjoint operators

Definition 9.11 (Self-adjoint). $T \in \mathcal{L}(V)$ is *self-adjoint* (*Hermitian*) if $T^\dagger = T$, i.e. $\langle Tu, v\rangle = \langle u, Tv\rangle$ for all u, v. Its matrix in an orthonormal basis is a Hermitian matrix, $A^\dagger = A$.

These are the observables of quantum mechanics. Two facts make them so.

Theorem 9.12 (Real eigenvalues). Every eigenvalue of a self-adjoint operator is real.

Proof. Let $Tv = \lambda v$ with $v \neq 0$. Then $\lambda \|v\|^2 = \langle v, Tv\rangle = \langle Tv, v\rangle = \overline{\lambda}\,\|v\|^2$, using self-adjointness in the middle step. Since $\|v\|^2 \neq 0$, $\lambda = \overline{\lambda}$ is real. ■

Both of the next two theorems turn on one small fact, which is worth isolating rather than smuggling into a proof.

Lemma 9.13 (A vanishing quadratic form). Let $S \in \mathcal{L}(V)$ satisfy $\langle v, Sv\rangle = 0$ for every $v \in V$. If $\mathbb{F} = \mathbb{C}$, then $S = 0$. If $\mathbb{F} = \mathbb{R}$, the same follows provided S is self-adjoint.

Proof. Expanding $0 = \langle u+v, S(u+v)\rangle$ and using $\langle u, S\rangle u = \langle v, S\rangle v = 0$ gives

$$\langle u, Sv\rangle = -\langle v, Su\rangle \qquad \text{for all } u, v. \tag{9.1}$$

Over $\mathbb{C}$, replace u by iu in (9.1). The left side becomes $\bar{i}\,\langle u, S\rangle v = -i\langle u, S\rangle v$ (conjugate-linear first slot) and the right side $-i\langle v, S\rangle u$ (linear second slot), so $\langle u, S\rangle v = \langle v, S\rangle u$. Together with (9.1) this forces $\langle u, S\rangle v = 0$ for all u, v, hence $Sv = 0$ for every v. Over $\mathbb{R}$ with S self-adjoint, $\langle v, S\rangle u = \langle Sv, u\rangle = \langle u, Sv\rangle$, and (9.1) reads $\langle u, S\rangle v = -\langle u, S\rangle v$, giving the same conclusion. ■

Theorem 9.14 (Self-adjoint via a real quadratic form). For $\mathbb{F} = \mathbb{C}$, an operator T is self-adjoint if and only if $\langle v, Tv\rangle \in \mathbb{R}$ for every v.

Proof. If $T^\dagger = T$ then $\langle v, Tv\rangle = \langle Tv, v\rangle = \overline{\langle v, Tv\rangle}$ is real. Conversely, if $\langle v, Tv\rangle$ is always real then $\langle v, Tv\rangle = \overline{\langle v, Tv\rangle} = \langle Tv, v\rangle = \langle v, T^\dagger v\rangle$, so $\langle v, (T - T^\dagger)v\rangle = 0$ for all v, so $T = T^\dagger$ by Lemma 9.13. ■

Theorem 9.15 (Orthogonal eigenvectors). Eigenvectors of a self-adjoint operator for distinct eigenvalues are orthogonal.

Proof. Let $Tu = \alpha u$, $Tv = \beta v$ with $\alpha \neq \beta$ (both real). Then

$$\alpha\langle u, v\rangle = \langle Tu, v\rangle = \langle u, Tv\rangle = \beta\langle u, v\rangle,$$

using $\langle Tu, v\rangle = \overline{\alpha}\langle u, v\rangle = \alpha\langle u, v\rangle$ on the left. Thus $(\alpha - \beta)\langle u, v\rangle = 0$, forcing $\langle u, v\rangle = 0$. ■

Physically: a measurement of a self-adjoint observable can only return one of its (real) eigenvalues, and states with different definite values are automatically orthogonal—hence perfectly distinguishable.

Example 9.16 (The Pauli matrices are observables). Each Pauli matrix $\sigma_x = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, $\sigma_y = \left(\begin{smallmatrix} 0 & -i \\ i & 0 \end{smallmatrix}\right)$, $\sigma_z = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$ equals its own conjugate transpose, so all three are self-adjoint with real eigenvalues ± 1—the two outcomes of a spin measurement—and orthogonal eigenstates. ◁

Example 9.17 (Orthogonal eigenvectors in action). The Hermitian matrix $A = \left(\begin{smallmatrix} 2 & i \\ -i & 2 \end{smallmatrix}\right)$ has real eigenvalues $1, 3$ with eigenvectors $(1, i)$ and $(1, -i)$. Their inner product $\langle (1, i), (1, -i)\rangle = \overline{1} \cdot 1 + \overline{i} \cdot (-i) = 1 + (-i)(-i) = 1 - 1 = 0$ confirms Theorem 9.15. ◁

Example 9.18 (An orthonormal eigenbasis). Normalizing those eigenvectors gives the orthonormal pair $e_1 = \frac{1}{\sqrt{2}}(1, i)$, $e_2 = \frac{1}{\sqrt{2}}(1, -i)$, in which the operator becomes $\operatorname{diag}(1, 3)$. A self-adjoint operator is thus diagonal in an orthonormal basis—the spectral theorem of Lecture 10, previewed. ◁

Example 9.19 (A 3×3 self-adjoint operator). The real symmetric matrix $\left(\begin{smallmatrix} 14 & -13 & 8 \\ -13 & 14 & 8 \\ 8 & 8 & -7 \end{smallmatrix}\right)$ equals its transpose, hence is self-adjoint, with real eigenvalues $27, 9, -15$ and orthonormal eigenvectors $\frac{1}{\sqrt{2}}(1, -1, 0)$, $\frac{1}{\sqrt{3}}(1, 1, 1)$, $\frac{1}{\sqrt{6}}(1, 1, -2)$. In that orthonormal basis it is $\operatorname{diag}(27, 9, -15)$—real spectrum, orthogonal eigenstates, exactly as the theory predicts. ◁

Remark 9.20 (The real case). A real symmetric matrix is self-adjoint for the real dot product, so it too has real eigenvalues and an *orthogonal* eigenbasis—the real spectral theorem. This is why the principal axes of an ellipsoid, the moments of inertia of a rigid body, and the normal modes of a coupled system come out mutually perpendicular.

Example 9.21 (Diagonalizing σ_x). $\sigma_x = \left(\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right)$ has real eigenvalues ±1 with orthonormal eigenvectors $|\pm\rangle = \frac{1}{\sqrt{2}}(1, \pm1)$. In the unitary basis $U = (|+\rangle\ |-\rangle)$,

$$U^\dagger \sigma_x U = \begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix} = \sigma_z,$$

so the Hadamard change of basis turns σ_x into σ_z: a self-adjoint operator becomes a real diagonal matrix in its orthonormal eigenbasis. ◁

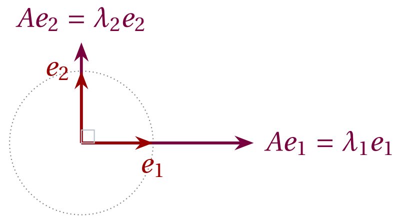

Figure 1: A self-adjoint operator stretches along an *orthogonal* eigenbasis $e_1 \perp e_2$ by real factors λ_1, λ_2; the unit circle maps to an ellipse with axes along the eigendirections.

Remark 9.22 (Products of observables). The product of two self-adjoint operators is self-adjoint if and only if they commute, since $(ST)^\dagger = T^\dagger S^\dagger = TS$. A product of observables is again an observable only for compatible (commuting) observables.

Example 9.23 (Expectation values are real). For a self-adjoint A and any state $|\psi\rangle$, the expectation value $\langle A\rangle = \langle\psi|\,A\,|\psi\rangle = \langle\psi, A\psi\rangle$ is real by Theorem 9.14. The measured average of an observable is therefore a real number, exactly as physics requires. ◁

Example 9.24 (Energies are real, eigenstates orthogonal). A Hamiltonian $\hat{H}$ is self-adjoint, so its energy eigenvalues E_n are real and eigenstates for distinct energies are orthogonal, $\langle E_m \mid E_n\rangle = \delta_{mn}$. These are the very facts that let us expand any state in the energy basis and evolve each term by a phase $e^{-iE_n t/\hbar}$ (Lecture 6). ◁

Example 9.25 (A real quadratic form). For $\sigma_z = \mathrm{diag}(1, -1)$ and $v = (a, b)$, $\langle v, \sigma_z v\rangle = \overline{a}\,a - \overline{b}\,b = |a|^2 - |b|^2 \in \mathbb{R}$, illustrating Theorem 9.14: the quadratic form of an observable is always real, whatever the state. ◁

3 Unitary operators

Definition 9.26 (Unitary). $U \in \mathcal{L}(V)$ is *unitary* if $U^\dagger U = UU^\dagger = I$, i.e. $U^\dagger = U^{-1}$.

Theorem 9.27 (Unitary means inner-product preserving). For $U \in \mathcal{L}(V)$ the following are equivalent:

1. U is unitary;
2. $\langle Uu, Uv\rangle = \langle u, v\rangle$ for all u, v (preserves the inner product);
3. $\|Uv\| = \|v\|$ for all v (preserves norms);

4. U maps some (every) orthonormal basis to an orthonormal basis.

Proof. (1)⇒(2): $\langle Uu, Uv\rangle = \langle u, U^\dagger Uv\rangle = \langle u, v\rangle$. (2)⇒(3): take $u = v$. (3)⇒(2): recover the inner product from the norm by polarization (Lecture 7). (2)⇒(4): if $\langle e_j, e_k\rangle = \delta_{jk}$ then $\langle Ue_j, Ue_k\rangle = \delta_{jk}$. (4)⇒(1): if U sends the ONB e_k to an ONB, then $U^\dagger U$ has matrix δ_{jk}, so $U^\dagger U = I$, and in finite dimensions a left inverse is a two-sided inverse. ∎

Theorem 9.28 (Unit-circle spectrum). Every eigenvalue of a unitary operator has absolute value 1.

Proof. If $Uv = \lambda v$ with $v \neq 0$, then $\|v\| = \|Uv\| = |\lambda|\,\|v\|$, so $|\lambda| = 1$; the eigenvalues lie on the unit circle, $\lambda = e^{i\theta}$. ∎

Time evolution $e^{-i\hat{H}t/\hbar}$ is unitary, which is why it preserves total probability $\|\psi\| = 1$; symmetry operations (rotations, reflections, parity) are unitary for the same reason.

Example 9.29 (Unitary operators). The rotation $R = \left(\begin{smallmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{smallmatrix}\right)$ satisfies $R^\dagger R = I$ and preserves lengths in $\mathbb{R}^2$. Each Pauli matrix is also unitary (besides being Hermitian), since $\sigma^\dagger\sigma = \sigma^2 = I$; indeed a *Hermitian* operator is unitary precisely when $A^2 = I$. (Self-inverse alone is not enough: $\left(\begin{smallmatrix} 1 & 1 \\ 0 & -1 \end{smallmatrix}\right)$ squares to I without being Hermitian.) ◁

Example 9.30 (Unit-circle eigenvalues). Over $\mathbb{C}$ the rotation R above has eigenvalues $e^{\pm i\theta}$ with eigenvectors $(1, \mp i)$: both lie on the unit circle, as Theorem 9.28 requires, and the eigenvectors are orthogonal because R is normal. ◁

Example 9.31 (Verifying unitarity). The Hadamard operator $U = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$ satisfies $U^\dagger U = \frac{1}{2}\left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)\left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right) = I$, so it is unitary; it maps the basis $|0\rangle, |1\rangle$ to the orthonormal pair $|\pm\rangle$ of Lecture 7. ◁

Example 9.32 (Unitarity preserves overlaps). Under the same Hadamard U, $|0\rangle \mapsto |+\rangle$ and $|1\rangle \mapsto |-\rangle$, and $\langle U0, U1\rangle = \langle + \mid - \rangle = 0 = \langle 0 \mid 1\rangle$: the overlap of the images equals the overlap of the originals, as Theorem 9.27 guarantees. ◁

Remark 9.33 (Change of orthonormal basis is unitary). The matrix relating two orthonormal bases has orthonormal columns, hence is unitary (Theorem 9.27). Passing between orthonormal bases is always a unitary transformation, and an operator's matrix transforms as $A' = U^\dagger AU$ with U unitary—the inner-product-respecting refinement of the similarity $A' = P^{-1}AP$ from Lecture 4.

Remark 9.34 (Generating unitaries). If A is self-adjoint then e^{iA} is unitary, because $(e^{iA})^\dagger = e^{-iA^\dagger} = e^{-iA} = (e^{iA})^{-1}$. This is the finite-dimensional shadow of Stone's theorem: every one-parameter unitary group e^{-iAt} is generated by a self-adjoint A—the Hamiltonian, when $A = \hat{H}/\hbar$ (Lecture 6).

4 Normal operators

Definition 9.35 (Normal). $T \in \mathcal{L}(V)$ is *normal* if it commutes with its adjoint, $T^\dagger T = TT^\dagger$.

Self-adjoint operators ($T^\dagger = T$) and unitary operators ($T^\dagger = T^{-1}$) are obviously normal; normality is the common umbrella. Its usefulness rests on one identity.

Theorem 9.36 (Normal length identity). T is normal if and only if $\|Tv\| = \|T^\dagger v\|$ for all v.

Proof. For any T, $\|Tv\|^2 = \langle Tv, Tv\rangle = \langle v, T^\dagger T v\rangle$ and $\|T^\dagger v\|^2 = \langle v, TT^\dagger v\rangle$. These agree for all v iff $\langle v, (T^\dagger T - TT^\dagger)v\rangle = 0$ for all v; as $T^\dagger T - TT^\dagger$ is self-adjoint, Lemma 9.13 says that holds iff it is 0. ■

Corollary 9.37 (Shared eigenvectors with the adjoint). If T is normal, then $Tv = \lambda v$ if and only if $T^\dagger v = \overline{\lambda} v$. Consequently eigenvectors of T for distinct eigenvalues are orthogonal.

Proof. $T - \lambda I$ is normal (it commutes with its adjoint $T^\dagger - \overline{\lambda} I$), so by Theorem 9.36 $\|(T - \lambda I)v\| = \|(T^\dagger - \overline{\lambda} I)v\|$; the left side is 0 iff the right side is. For orthogonality, with $Tu = \alpha u$ and $Tv = \beta v$ we get $T^\dagger v = \overline{\beta} v$, so $\overline{\alpha}\langle u, v\rangle = \langle Tu, v\rangle = \langle u, T^\dagger v\rangle = \overline{\beta}\langle u, v\rangle$, forcing $\langle u, v\rangle = 0$ when $\alpha \neq \beta$. ■

Example 9.38 (Normal but neither Hermitian nor unitary). $A = \left(\begin{smallmatrix} 1 & 1 \\ -1 & 1 \end{smallmatrix}\right)$ has $A^\dagger = \left(\begin{smallmatrix} 1 & -1 \\ 1 & 1 \end{smallmatrix}\right)$ and $AA^\dagger = A^\dagger A = 2I$, so A is normal; yet $A^\dagger \neq A$ and $AA^\dagger = 2I \neq I$, so it is neither self-adjoint nor unitary. Its eigenvalues $1 \pm i$ are neither real nor on the unit circle—the general normal case. ◁

Example 9.39 (Checking normality by lengths). For the same A, $\|Av\|^2 = \langle v, A^\dagger A v\rangle = 2\|v\|^2 = \langle v, AA^\dagger v\rangle = \|A^\dagger v\|^2$, so $\|Av\| = \|A^\dagger v\|$ for every v—the length test of Theorem 9.36, passed. ◁

Remark 9.40 (Positive operators). For any T, the operator $T^\dagger T$ is self-adjoint and *positive*: $\langle v, T^\dagger T v\rangle = \|Tv\|^2 \geq 0$, so its eigenvalues are nonnegative. Positive operators are the analogue of the nonnegative reals and underlie the singular value decomposition and quantum density matrices.

Example 9.41 (Orthogonal eigenvectors of a normal operator). The normal operator $A = \left(\begin{smallmatrix} 1 & 1 \\ -1 & 1 \end{smallmatrix}\right)$ has eigenvalues $1 \pm i$ with eigenvectors $(1, \pm i)$, and $\langle (1, i), (1, -i)\rangle = 1 + (-i)(-i) = 0$. Distinct eigenvalues give orthogonal eigenvectors, as Corollary 9.37 promises—even though A is neither self-adjoint nor unitary. ◁

Remark 9.42 (Polar and singular value decompositions). Just as every complex number is $re^{i\theta}$, every operator factors as $T = UP$ with U unitary and $P = \sqrt{T^\dagger T}$ positive (the *polar decomposition*); diagonalizing P in orthonormal bases yields the *singular value decomposition* $T = U\Sigma V^\dagger$, the workhorse of data analysis and low-rank approximation. Both rest on the adjoint built here.

Theorem 9.43 (Cartesian decomposition). Every operator on a complex space is $T = A + iB$ with $A = \frac{1}{2}(T + T^\dagger)$ and $B = \frac{1}{2i}(T - T^\dagger)$ self-adjoint, and T is normal if and only if A and B commute.

Proof. Both A and B are self-adjoint by direct check, and $A + iB = T$. A short computation gives $AB - BA = \frac{1}{2i}(T^\dagger T - TT^\dagger)$, which vanishes exactly when T is normal. The pair (A, B) are the operator "real and imaginary parts" of T, in analogy with $z = a + ib$. ■

Example 9.44 (Cartesian parts of an operator). For $T = \left(\begin{smallmatrix} 1 & 2i \\ 0 & 3 \end{smallmatrix}\right)$, $T^\dagger = \left(\begin{smallmatrix} 1 & 0 \\ -2i & 3 \end{smallmatrix}\right)$, so the self-adjoint parts are

$$A = \frac{T + T^\dagger}{2} = \begin{pmatrix} 1 & i \\ -i & 3 \end{pmatrix}, \qquad B = \frac{T - T^\dagger}{2i} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

both Hermitian, with $T = A + iB$. Here $AB \neq BA$, so T is not normal—as expected for a non-diagonal upper-triangular matrix. ◁

Remark 9.45 (Skew-adjoint operators). An operator with $B^\dagger = -B$ is *skew-adjoint*; then iB is self-adjoint, so skew-adjoint operators have purely imaginary eigenvalues. Their exponentials are unitary, which is why the generator of time evolution, $-i\hat{H}/\hbar$, is skew-adjoint—the bridge between observables and symmetries.

Example 9.46 (A skew-adjoint operator). $B = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ satisfies $B^\dagger = -B$, so it is skew-adjoint with purely imaginary eigenvalues $\pm i$; indeed $iB = \left(\begin{smallmatrix} 0 & -i \\ i & 0 \end{smallmatrix}\right) = \sigma_y$ is Hermitian. Its exponential $e^{\theta B}$ is precisely the rotation of Example 9.29: a unitary generated by a skew-adjoint operator. ◁

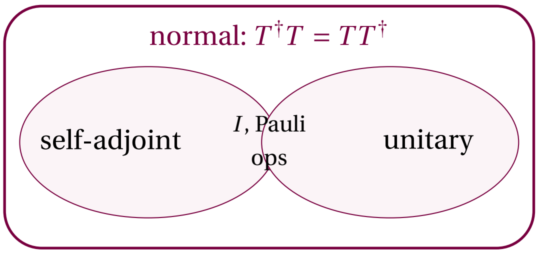

Figure 2: The self-adjoint and unitary operators are distinct families inside the normal operators; their overlap (operators that are both, such as I and the Pauli matrices) consists of self-inverse Hermitian operators.

5 Observables, symmetries, and what comes next

The dictionary is now complete. A self-adjoint operator plays the role of a real number: its eigenvalues are real (measurement outcomes) and its eigenvectors for distinct outcomes are orthogonal (distinguishable states). A unitary operator plays the role of a complex number of modulus one: it preserves all inner products and its eigenvalues are phases $e^{i\theta}$. Both are normal, and normality is exactly the property—by the spectral theorem of Lecture 10—that guarantees an orthonormal basis of eigenvectors.

This is why quantum mechanics works as it does: observables are self-adjoint, so they have a complete orthonormal set of eigenstates with real eigenvalues; time evolution and symmetries are unitary, so they shuffle these states without disturbing probabilities. The next lecture proves that every normal operator on a complex space is diagonal in an orthonormal basis, the cleanest possible form of an observable.

Proposition 9.47 (Uncertainty relation). For self-adjoint A, B and a unit state $|\psi\rangle$, writing $\Delta A = A - \langle A\rangle\, I$,

$$\langle(\Delta A)^2\rangle\,\langle(\Delta B)^2\rangle \geq \tfrac{1}{4}|\langle[A,B]\rangle|^2.$$

Proof. $\Delta A, \Delta B$ are self-adjoint, so $\langle(\Delta A)^2\rangle = \|\Delta A|\psi\rangle\|^2$. Cauchy–Schwarz gives

$$\|\Delta A\psi\|^2\,\|\Delta B\psi\|^2 \geq |\langle\Delta A\psi, \Delta B\psi\rangle|^2.$$

The imaginary part of $\langle\Delta A\psi, \Delta B\psi\rangle$ equals

$$\tfrac{1}{2i}\big(\langle\Delta A\psi, \Delta B\psi\rangle - \langle\Delta B\psi, \Delta A\psi\rangle\big) = \tfrac{1}{2i}\langle\psi|\,[A,B]\,|\psi\rangle,$$

using $[\Delta A, \Delta B] = [A, B]$. Since $|z|^2 \geq (\operatorname{Im} z)^2$, the bound follows. ■

Example 9.48 (Incompatible spin components). The Pauli matrices satisfy $[\sigma_x, \sigma_y] = 2i\sigma_z$, so the bound reads $\langle(\Delta\sigma_x)^2\rangle\langle(\Delta\sigma_y)^2\rangle \geq |\langle\sigma_z\rangle|^2$: two spin components cannot both be sharp unless the third has zero expectation. This is Heisenberg uncertainty in its barest form. ◁

Remark 9.49 (Compatible observables). Commuting self-adjoint operators can be simultaneously diagonalized in a single orthonormal basis (Lecture 10). Physically, *compatible* observables share a complete set of eigenstates and can hold definite values together; incompatible ones such as σ_x and σ_z cannot, which is the origin of the uncertainty relation.

Summary of main concepts

The adjoint $T^\dagger$, defined by $\langle T^\dagger u, v\rangle = \langle u, Tv\rangle$, is the conjugate transpose in an orthonormal basis and obeys $(ST)^\dagger = T^\dagger S^\dagger$, $(T^\dagger)^\dagger = T$, with $\operatorname{null} T^\dagger = (\operatorname{range} T)^\perp$. Self-adjoint operators ($T^\dagger = T$) have real eigenvalues and orthogonal eigenvectors—the observables. Unitary operators ($U^\dagger = U^{-1}$) preserve inner products and norms, map orthonormal bases to orthonormal bases, and have eigenvalues on the unit circle—the symmetries. Normal operators ($T^\dagger T = TT^\dagger$) satisfy $\|Tv\| = \|T^\dagger v\|$, share eigenvectors with $T^\dagger$ (conjugate eigenvalues), and have orthogonal eigenvectors for distinct eigenvalues; they are the operators the spectral theorem diagonalizes in an orthonormal basis.

- **Adjoint**—$\langle T^\dagger u, v\rangle = \langle u, Tv\rangle$; the conjugate transpose in an orthonormal basis; $\operatorname{null} T^\dagger = (\operatorname{range} T)^\perp$.
- **Self-adjoint ($T^\dagger = T$)**—the observables; real eigenvalues and orthogonal eigenvectors.
- **Unitary ($U^\dagger = U^{-1}$)**—the symmetries; preserve inner products, spectrum on the unit circle.
- **Normal** ($T^\dagger T = TT^\dagger$)—commutes with its adjoint, so $\|Tv\| = \|T^\dagger v\|$; orthogonal eigenvectors for distinct eigenvalues, the spectral-theorem class.

Exercises for this lecture are in the companion sheet `exercises-9.pdf`.